Course Project

1. Overview

The course project is intended to give you experience with the complete machine-learning workflow: defining a problem, understanding and auditing data, constructing appropriate baselines, designing experiments, evaluating models, analyzing failures, and communicating results.

You should be able to explain:

- why a particular modeling approach is appropriate
- what a more sophisticated model improves over a simpler alternative
- where the model fails and why
- whether its confidence can be trusted
- how sensitive the conclusions are to the train/test split and evaluation design
- and whether the additional computational complexity is justified by the improvement obtained.

The project will proceed through:

Proposal → Milestones → Poster → Final submission

You may follow one of two tracks.

Track A — Instructor-Proposed Projects

You choose one of the datasets and problems described in Section 4.

Track B — Student-Proposed Project

You may propose their own research-based, application-oriented, or theoretical project. The proposal must identify a meaningful machine-learning question and an appropriate evaluation methodology.

Track B projects require instructor approval.

2. Groups

Each group should be composed of two to three students. Students who do not form groups will be assigned into groups randomly. Groups should be formed within the same course section to facilitate coordination and project supervision.

All students in a group are expected to understand and be able to explain the complete project, not only the component they personally implemented.

3. Common Project Requirements

The emphasis of the project is not solely on achieving the highest possible predictive performance. Projects will be evaluated primarily on the quality of the experimental methodology, reasoning, analysis, interpretation, and conclusions. A well-designed experiment that produces a negative result is still a valid and valuable result. Where appropriate, you should also consider model interpretability and whether gains in predictive performance justify reductions in interpretability or increases in complexity.

3.1 The Model Ladder

Every empirical project should contain at least two levels of comparison.

Level 1 — Strong Classical Baseline

Every project should implement and carefully tune at least one conventional machine-learning approach appropriate to the task.

Examples include:

- logistic or linear regression
- decision trees
- random forests
- gradient-boosted trees
- nearest-neighbor methods
- or other classical statistical-learning approaches.

Feature construction, preprocessing, regularization, hyperparameter selection, and validation should be treated as meaningful modeling decisions.

Level 2 — More Sophisticated Method

You should investigate at least one method that they reasonably expect to improve upon the classical baseline.

This might involve:

- neural networks
- pretrained models
- or another justified extension.

You must explicitly answer:

What does this method improve over the simpler model, and what does that improvement cost?

3.2 Failure Analysis

Failure analysis is a required component of every project. You should not merely report aggregate test metrics and list a few misclassified examples. They should actively attempt to determine where the system stops working.

For important failure modes, you should propose and, where practical, evaluate a mitigation.

3.3 Complexity Analysis

A recurring question across the projects will be:

When is additional model or automation complexity actually worth using?

The purpose is not to argue that classical methods are universally preferable, nor that modern foundation models, Large Language Models, or agentic approaches are universally preferable.

Instead, you should learn to measure the trade-off.

4. Track A Projects

4.1 TABULAR — NASA Kepler Objects of Interest

Exoplanet Candidate Classification

The NASA Kepler dataset contains information about thousands of objects observed by the Kepler Space Telescope while searching for planets outside our solar system. Each object is described using measurements related to the possible planet, its orbit, the observed transit signal, and the star that it may be orbiting.
Your task is to build a machine-learning model that can distinguish between potential exoplanets and false positive detections. The aim is to investigate whether machine learning can help scientists identify the most promising observations for further study.

An important part of this project is deciding which variables should be used by the model. Some fields in the dataset may contain information that was produced after scientists had already investigated or classified an object. Using these variables could make the model appear much more accurate than it really is.
In addition to reporting model performance, you should investigate where the model makes mistakes, such as weak signals, unusual stars, missing measurements, or false positives that closely resemble real planets.

4.2 NATURAL LANGUAGE — Amazon Shopping Queries

Product Relevance in Online Shopping

The Amazon Shopping Queries dataset contains real customer search queries together with products that may or may not be relevant to those searches. For each product, the dataset includes information such as its title, description, bullet points, brand, and other text-based details. Each query–product pair

is labelled as one of four categories: Exact, Substitute, Complement, or Irrelevant.
Your task is to build a model that can predict how well a product matches a customer's search query. The aim is to investigate how effectively different natural-language methods can understand shopping intent and product relevance.

In addition to reporting overall performance, you should investigate where the model makes mistakes. Examples might include products that share many words with the query but are not actually relevant, useful products that use different wording from the search query, misspellings, unusual or rare queries, and cases where a more advanced language model still fails

4.3 IMAGE — VinDr-CXR

Identifying Abnormalities in Chest X-Rays

The VinDr-CXR dataset contains 18,000 chest X-ray images that have been reviewed by expert radiologists. The dataset includes 15,000 training images and 3,000 test images. The radiologists recorded a range of possible abnormalities in the X-rays. For some findings, they marked the location of the abnormality using a bounding box, while other diagnoses are provided as labels for the whole image. An important feature of this dataset is that several radiologists independently reviewed the same training images, so they do not always agree about what is visible.

Your task is to build an image-based machine-learning model that can identify abnormalities in chest X-rays. An important part of this project is deciding how to handle disagreement between radiologists. Medical images are not always easy to interpret, and two experts may reasonably reach different conclusions about the same X-ray. You should use an appropriate detection or localization measure. You may also investigate whether the model's confidence is reliable, particularly for difficult or uncertain cases.
In addition to reporting overall performance, you should investigate where the model makes mistakes. For example, you might examine whether subtle abnormalities are harder to detect, whether the model becomes less confident when radiologists disagree, and whether changes in image quality affect performance.

5. Deliverables and Milestones

Deliverable	Suggested Weight
Grouping and Topic Selection	0%
Poster and Presentation	20%
Final Report	60%
Code and Reproducibility	20%
Total	**100%**

Deadlines will be announced separately.

5.1 Project Milestones

1. **Project Proposal and Initial Baseline** — October 7
2. **Progress Report and Preliminary Results** — November 6
3. **Final Submission** — November 20

6. Poster

At the end of the project, each group should prepare a poster that provides a clear and concise summary of the problem addressed, the methodology used, and the main findings of the project. One poster per group should be submitted. The poster should communicate the following components:

1. Research Question: Clearly state the problem or research question addressed by the project. Briefly explain why the problem is important or interesting.
2. Introduction: describe your problem in detail and summarize your approach and findings.
3. Dataset and Evaluation: Describe the dataset used, including its main characteristics and any relevant preprocessing. Summarize the experimental setup and evaluation metrics used to assess model performance.
4. Methods: describe the main method, model, or approach developed or investigated in the project. Highlight the most important differences between the proposed approach and the baseline.
5. Results: present the most important quantitative result of the project. Whenever possible, use a table, figure, or plot that clearly compares the classical baseline with the more sophisticated method.
6. Conclusion: clearly state the main conclusion that can be drawn from the project. The conclusion should be supported by the experimental results presented on the poster.
7. References: a list of references that you used throughout your project.

Your poster will be graded based on the following rubric:

7. Final Report

At the end of the project, you should create a report describing how you tackled your chosen problem. The report should consist of the below sections. One report per group should be submitted. You are encouraged to use latex for typing. The report should include:

1. Abstract: provide a summary describing the problem you chose, how you tackled it, and a summary of your results.
2. Introduction: describe your problem in detail and summarize your approach and findings.
3. Dataset: describe the dataset you used to train your models, and any pre-processing steps you performed.
4. Models: describe the models you used to tackle your problem, and how you trained, tuned and validated them. Please be specific and concise. Also, provide the rationale behind your choices.

5. Results: describe the experiments you ran to train and evaluate your models. Also, provide all results of your experiments and discuss them. Such experiments include complexity and resource analysis, as well as failure analysis.
6. Conclusion: provide a concise conclusion of your project, highlighting the main results you achieved, and what future work could be.
7. References: a list of references that you used throughout your project.

Your report will be graded based on the following rubric:

8. Code and Reproducibility

You should submit a notebook that includes all the code you used to tackle your problem including data cleaning and data processing, training, diagnosis, hyper-parameter tuning and model selection. You notebook should be clear, well-structured, and well commented. In addition to submitting the notebook through Moodle, you are required to maintain a Github repository with a properly documented code and include the link of that repository in your report. The Github repository should have your final trained model available for download. You are responsible for understanding, testing, debugging, and defending all submitted code. Your code will be graded based on the following rubric: